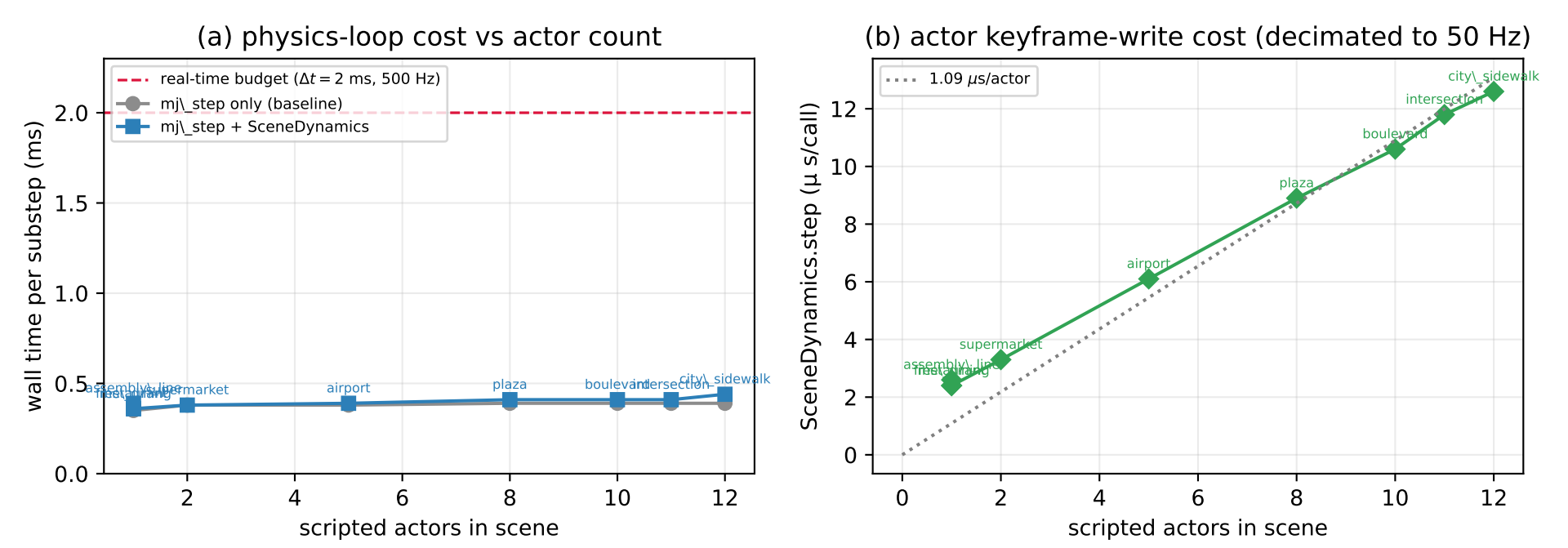
(a) physics-loop cost vs actor count
real-time budget (Δt = 2 ms, 500 Hz)
mj_step only (baseline)
mj_step + SceneDynamics
wall time per substep (ms)
scripted actors in scene
0.0
0.5
1.0
1.5
2.0
2
4
6
8
10
12
assembly_line
supermarket
airport
plaza
boulevard
intersection
city_sidewalk
(b) actor keyframe-write cost (decimated to 50 Hz)
1.09 μs/actor
SceneDynamics.step (μ s/call)
scripted actors in scene
0
2
4
6
8
10
12
supermarket
airport
plaza
boulevard
intersection
city_sidewalk